

(W)
Enter WASD (or QUIT): (A) (S) (D)
--snip--

How It Works

The data structure that represents the sliding tile game board is a list of lists. The inner lists each represent one column of the 4 × 4 board and contain strings for the numbered tiles (or the BLANK string to represent the blank space). The `getNewBoard()` function returns this list of lists with all tiles in their starting positions and the blank space in the lower-right corner.

Python can swap the values in two variables with a statement like `a, b = b, a`. The program uses this technique on lines 101 to 108 to swap the blank space and a neighboring tile and simulate sliding a numbered tile into the blank space. The `getNewPuzzle()` function generates new puzzles by performing 200 of these swaps randomly.

```
1. """Sliding Tile Puzzle, by Al Sweigart al@inventwithpython.com
2. Slide the numbered tiles into the correct order.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: large, game, puzzle"""
5.
6. import random, sys
7.
8. BLANK = '  ' # Note: This string is two spaces, not one.
9.
10.
11. def main():
12.     print('''Sliding Tile Puzzle, by Al Sweigart al@inventwithpython.com
13.
14.     Use the WASD keys to move the tiles
15.     back into their original order:
16.         1  2  3  4
17.         5  6  7  8
18.         9 10 11 12
19.        13 14 15  ''')
20.     input('Press Enter to begin...')
21.
22.     gameBoard = getNewPuzzle()
23.
24.     while True:
25.         displayBoard(gameBoard)
26.         playerMove = askForPlayerMove(gameBoard)
27.         makeMove(gameBoard, playerMove)
28.
29.         if gameBoard == getNewBoard():
30.             print('You won!')
31.             sys.exit()
32.
33.
34. def getNewBoard():
35.     """Return a list of lists that represents a new tile puzzle."""
```